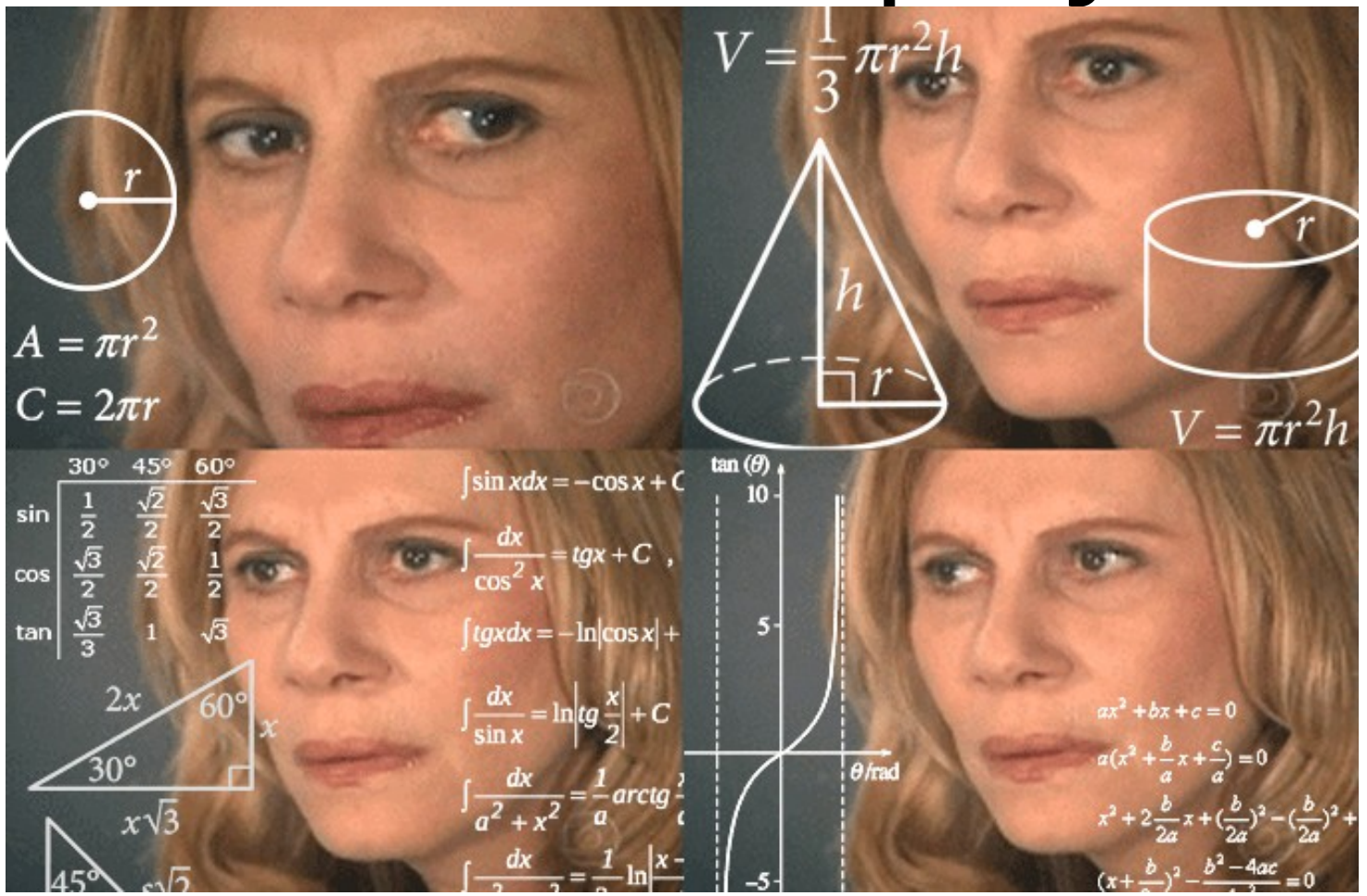
Science Inquiry



Data and Analysing Results

Quantifying Results

- Qualitative data:
  - observations that do not involve measurement
  - Eg: “Several respondents reported feeling better after receiving the medication.”

- Quantitative data:
  - Observations with a numerical value, that can be measured
  - Eg: “On average, the experimental group experienced a 20 point drop in systolic blood pressure after receiving the medication.

Try to get quantitative data where possible

Errors in Data

- Error: in some cases, mistakes, in others:

  “a deviation from the result that would have been obtained under perfect conditions”.

- Variations in sample and/or measurement process can cause “measurement error”.
  - Not necessarily a “mistake”, may be due to limitations of equipment
  - Large sample sizes can help.

Limitations of Data

- Can only draw conclusions that are supported by the data.
- Beware of other, uncontrolled factors that can affect the data.
- Media reporting often over-simplifies research data, drawing misleading conclusions.

Secondary Data

- Data collected by researchers, then re-used or re-analysed by other researchers: “second hand”

Confidence Interval

- Used to indicate reliability of data.
- Range of values in which the true value is likely to occur, within a margin of error.

Eg: If a study shows that 53% of people plus or minus 1.5% experience a drop in blood pressure after taking *Pressurego*, the confidence interval is 51.5-54.5%

- Used alongside Confidence Level

Confidence Level

- Used alongside Confidence Interval
- Calculated using statistical analsysi.
- The % of cases which fall within the confidence interval

Eg: a 95% confidence level means that, if you repeat the experiment multiple times, the range of values you obtain will fall within the confidence interval 95% of the time.

So, if you complete the *Pressurego* experiment multiple times, the proportion of people who experience a drop in blood pressure will be between 51.5 and 54.5, 95% of the time.

- A greater than 95% confidence level indicates that the results are reliable, and is considered the minimum acceptable confidence level in research.

Processing Data

- **Mean:**
  - Add values, then divide by number of values.
  - Can be strongly affected by outliers.
  - Can remove outlier if very clearly resulting from error.
- **Median:**
  - Middle value in a set of numbers
  - Reduces effect of outliers
  - Useful for some types of data

- **Range:**
  - Difference between largest and smallest value
  - Gives an idea of how “clustered together” results are.
  - In formal scientific research, range can be measured as standard deviation or quartiles.

- **Ratios and Rates**
  - Numerical statement of how one variable relates to another.

- **Percentages:**
  - Used to express how large part of something is.
  - /100
- **Percentage Change:**
  - Good way to show change over time.
  - Negative = decrease
  - Positive = increase
- **Frequency:**
  - Number of times an event occurs
- **Probability:**
  - The chance that a particular event will occur.

Percent Change

$$\text{Percent Change} = \frac{\text{New Value} - \text{Old Value}}{\text{Old Value}} \times 100\%$$

If the result is positive, it is an increase.
If the result is negative, it is a decrease.

Presentation of Data

- **Tables:**
  - Title, stating variables
  - Columns, for IV and DV
  - Headings on columns, including units
- **Graphs:**
  - Pictorial way of presenting data
  - Title, labelled axes, units, correct plotting

- **Models:**

A simplified representation of an idea or process, used to predict what would happen.

- **Flow Charts**

A diagram showing steps involved in a process.